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Dalhousie University

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Department of Psychology

22nd Annual In-House Convention

29-30 April 1996

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SCHEDULE

Monday, April 29, 1996 (LSC 4258/63)

9:00 - 9:15 Coffee

9:15 - 9:30 Opening Remarks—Richard Brown

9:30 - 11:00 Session 1. Chair: Tara Fidler
T1) R.E. Brown, T. Moran and B. Slotnick
poster P6 ~~T2)~~ R.E. Brown, F. Dastur, A. McLellan and I. MacGregor
T3) R.E. Brown and D. Cantoni
X? ~~T4) K. Firth, P. Neumann and R. Brown~~
T5) C. Forestall
T6) D. Santor and Z.V. Segal

Paul
Bill

11:00 - 11:15 Coffee Break

11:15 - 12:30 Session 2. Chair: J. Barresi
T7) D. Tripp and M. Sullivan
T8) H. MacLatchy-Gaudet
T9) S. Jerrott, S. Stewart and S. Dunphy
T10) P. McGrath
T11) S. Symons, H. MacLatchy-Gaudet and J. Milner

12:30 - 2:00 Lunch Break and Posters (Lounge):
P1) J. Christie
X ~~P2) W.C. Schmidt~~ Talk 20
P3) D. Symons

2:00 - 3:30 Session 3. Chair: B. Rusak
T12) J. Barresi
T13) S. Ling and S. Nakajima
T14) T. Fidler and V. LoLordo
T15) M. Yoon and J. Connolly
T16) B. Losier, P. McGrath and R. Klein

3:30- 3:45 Coffee Break

3:45- 5:00 Session 4. Chair: V. LoLordo
T17) A. MacDonald
X ~~T18) K. MacKinnon~~
T19) Y. Gidron
T20) W.C. Schmidt
T20a) Anne Marie Brown-Dequire (was P5)

There will be a cash bar social in the Lounge after the last session of the day.

Tuesday, April 30, 1996 (LSC 4258/63)

9:00 - 9:15 Coffee

9:15 - 10:45 Session 5. Chair: J. Connolly
T21) D. Symons, E. McLaughlin and L. Noiles
T22) C. Moore
T23) K. Skene
T24) P. Mendella
T25) R. Hoffman
T26) W.C. Schmidt

10:45 - 11:00 Coffee Break

11:00 - 12:30 Session 6. Chair: S. Adamo
T27) T.L. Taylor, D.P. Phillips, S.E. Hall, M.M. Carr and J.E. Mossop
T28) D.P. Phillips, T.L. Taylor and S.E. Hall
T29) J. Hamm and P. McMullen
T30) S. Simpson and P. McMullen
T31) S. McColl

12:30 - 2:00 Lunch Break and Posters (Lounge):
P4) D. Voyer, J. McGlone and J. Savoie
P5) ~~A. M. Brown-DeGagne,~~ J. McGlone and D. Santor See Session 4,

2:00 - 3:30 Session 6. Chair: W.C. Schmidt
T32) S.Adamo
T33) P. Campagnaro, G. Scott and B. Rusak
T34) S. Grewal
T35) M. Guido, D. Goguen, B. Rusak and H. Robertson
P5 ~~T36)~~ S.-W. Ying, B. Rusak, P. Delagrange, E. Mocaër, P. Renard and B. Guardiola-Lemâitre

3:30 - 3:45 Coffee Break

3:45 - 4:45 Symposium "How Clinical and Experimental Psychologists Should Work Together: Towards a Scientist/Practitioner Model"

- S. Stewart
- J. McGlone
- R. Klein
- G. Eskes
- D. Santor
- P. McMullen
- S. McColl
- F. Dastur
- B. Joyce (N.S. Rehabilitation Centre)

4:45 - 5:00 Closing Remarks and Awards

There will be pizza and beer available in the Lounge after the last session of the day.

All talk sessions will be held in the LSC Rooms 4258/63.

T1. What a Rat's Nose Knows

R.E. Brown, T. Moran & B. Slotnick

In experiment 1 an aqueous solution of amyl acetate was used as an odour cue for illness using a one-bottle choice test with CS-US intervals ranging between 15 minutes and 8 hours. The results showed that amyl acetate was an effective cue for illness with CS-US delays up to two hours. In experiment 2 the effects of the muscarinic antagonist scopolamine HBr on odour aversion learning was examined using both one and two bottle choice tests. Injection of scopolamine HBr (0.01 and 0.1 mg/kg) 30 minutes prior to the conditioning trial suppressed fluid intake on the conditioning trial, but did not disrupt acquisition of an illness-induced aversion to the amyl acetate solution. Injection of scopolamine HBr (0.1 mg/kg) 15 minutes after the conditioning trial did not suppress fluid intake and did not interfere with acquisition of an illness- induced odour aversion. Experiment 3 assessed the effects of the NMDA receptor antagonist MK-801 on acquisition of an odour aversion. MK-801 (0.05 mg/kg) did not interfere with acquisition of an odour aversion. Experiment 4 examined the effects of odour exposure only on acquisition of an illness odour aversion. The results demonstrated that fluid intake appears to be necessary for an illness-induced odour aversion to form. These results challenge the Sensory and Gate Channeling theory that only tastes are capable of acting as cues for illness in food aversion learning (Rusiniak, Hankins, Garcia & Brett, 1979). These experiments also demonstrated that glutamatergic and cholinergic neurotransmitter systems appear to play little or no role in acquisition of an illness-induced odour aversion.

explain methods better

PC ~~T2~~. The Cannabinoid Agonist CP55,940 Reduces Ultrasonic Vocalizations of Infant Rats

Poster

Richard Brown, Farhard Dastur, Ashley McLennan and Iain McGregor

We injected 12 day old rats with the cannabinoid agonist CP55,940 (10, 100 or 1000 mg/kg) or with combinations of CP55,940 and the cannabinoid antagonist SR141716A (20 mg/kg), the opioid antagonist naloxone (1.0 mg/kg), the dopaminergic D1 antagonist SCH23390 (0.5mg/kg) or the benzodiazepine antagonist flumazenil (20 mg/kg). CP55,940 dose-dependently reduced body temp., activity levels and the number of UVs emitted. These effects were antagonised by SR141716A, but not by SCH23390, naloxone or flumazenil These results suggest that CP55,940 has anxiolytic effects in rat pups through an action that is specific to the cannabinoid receptor.

T3. The Influence of the Male California Mouse (*Peromyscus californicus*) on Pup Survival and Female Reproductive Success

Richard E. Brown and Debora Cantoni

The California mouse is a monogamous rodent in which the male shows a high level of parental care. In our experiments, these mice were required to "forage" for food by running in a wheel. Male-female pairs could rear larger litters of pups to weaning than females alone and the pups with both parents grew faster. Single females who reared their pups to weaning had a longer inter-birth-interval than paired females. Our presentation will discuss the behaviour of the male which increases pup survival and facilitates implantation in the female.

T4. Sexual and Maternal Behavior Correlates of Neuroanatomical Differences Between Inbred Mouse Strains

Kara Firth, Paul Neumann and Richard Brown

In rodents, the preoptic area (POA) of the hypothalamus has been demonstrated to have critical involvement in sexual and maternal physiology and behaviors. Structural differences in the POA have been found between two inbred mouse strains, DBA/2J (D2) and C57BL/6J (B6) (Robinson et al, 1985). The B6 strain lacks a densely packed neuronal cluster (termed 'medioventral pars compacta') that is found in the D2 strain in the POA, on either side of the third ventricle; this compact nucleus may be homologous to The sexually dimorphic nucleus (SDN) in rats. Preliminary results (Mathieson and Neumann, unpublished) suggest that this hypothalamic structural variant is an X-linked trait.

These strains also show differences on sexual behavior measures such as male sexual performance (McGill and Manning, 1976) and maternal behavior measures such as infanticide (Svare and Broida, 1982; Svare et al, 1984). It is therefore possible that the structural variation between the two strains is responsible for the observed behavioral differences. Specific aims of this project are 1) to find experimental protocols that will reproduce published strain differences in sexual and maternal behaviors between D2 and B6 mice, and/or establish new ones, 2) to determine the mode of inheritance of these differences in classical crosses, and 3) to correlate variation in these behaviors with inheritance of the hypothalamic variant. Therefore, this project provides a unique opportunity to consider a model system involving genetic variation and the resulting structural differences between mouse strains, and the postulated behavioral manifestations of these differences.

T5. Roles of Tastes and Odours in Flavour Preference Conditioning

Cathy Forestall

use stronger odours,

In flavor aversion conditioning, tastes tend to facilitate or potentiate conditioning to odors when both stimuli are presented simultaneously. In the study to be discussed, the roles of tastes and odors in flavor preference conditioning were investigated using a reverse order reinforcement procedure. In the Experimental Condition, animals were exposed to an 8% sucrose solution followed after 5 minutes by either a citric acid solution (Taste Group) or an almond solution (Odor Group). The Taste and Odor Groups in the Control Condition received sucrose followed by either citric acid or almond solutions after 1 hour. During the test phase, there was a trend for animals in the Experimental Condition to prefer the taste or odor paired with the sucrose more than the Control animals, however this difference was significant in the Odor Group only on day 2 and in the Taste Group on day 3. Since stronger conditioning does not occur to tastes compared to odors when each is presented alone, overshadowing rather than potentiation may occur when odors and tastes are conditioned simultaneously.

T6. Predicting Relapse in Depression from Improvement Rates Achieved in Therapy

Darcy Santor and Zindel V. Segal

Studies examining treatment efficacy for psychotherapeutic interventions and pharmacological agents have focused primarily on the degree to which symptoms have been reduced or ameliorated over the course of the treatment protocol. Few studies have examined how individual differences in (a) the rate of change or (b) week-to-week variability are related to termination or follow-up symptom scores. In the present study, levels of depression were assessed weekly and again at a 3-month follow-up using the Beck Depression Inventory in a sample (N=106) of depressed outpatients receiving cognitive-behavioural therapy. Results showed that change is extremely variable across individuals (see Figure 1) and that depressive severity at the 3-month follow-up was strongly related to (a) the rate at which individuals improved during therapy and (b) the amount of week-to-week variability. Although all individuals receiving therapy had termination BDI scores within a non-clinical range, the rate at which individuals improved was crucial in determining 3-month follow-up BDI scores. Results suggest the need for identifying factors that influence not just the amount of change achieved in therapy but the manner in which that change is achieved. Failure to obtain smooth and steady improvement within initial sessions may represent a risk factor for relapse.

T7. Men, Women, and Pain-related Catastrophic Thinking

Dean Tripp and Michael Sullivan

The Pain Catastrophizing Scale (PCS; Sullivan, Bishop, & Pivak, 1995) is a significant predictor of self-reported pain measures for both clinical and nonclinical samples. Although Sullivan, Bishop, and Pivak (1995) found no significant effects of gender, data do indicate a trend with women reporting higher levels of catastrophizing. Due to small sample sizes and power, the individual differences issue is unresolved. Investigating gender differences, 799 Dalhousie students (Women: n=520, Men: n=279) completed the PCS. A principal components analysis with oblique rotation was conducted for each sex. Findings suggest that the women will cope with pain by engaging cognitions focused on their inability to deal with the pain (i.e., helplessness), while men are more focused with their inability to stop pain-related thoughts from occurring (i.e., rumination). In light of this data, Endler and Parker (1994), and Jensen, et al. (1994) must be considered to have valid points: there are gender differences in pain catastrophizing.

T8. Effects of Various Planning Prompts on Searching an Unfamiliar Text

Heather MacLatchy-Gaudet

The ability to search for and locate information in a textbook is a valuable literacy skill in university. It has been shown that search accuracy improves when students are prompted to plan their search, particularly if the question does not contain an explicitly stated search term. Also, the selection of appropriate categories of text to search is facilitated by prior knowledge of the topic area. This study was designed to assess the utility of three prompt sets (general, specific, or general + specific) on more and less difficult search tasks. It was hypothesized that specific prompts alone would be as facilitative of search as the general + specific prompts. Sixty introductory psychology students completed six search questions in an unfamiliar introductory psychology textbook, with 45 receiving one of the three prompt sets before each task. All had some level of prior knowledge, as they has completed as least one term of Introductory Psychology. No differences in accuracy, speed or category selection were found among the prompt conditions. Indexed quesitons were completed faster than non-indexed questions, with fewer pages viewed, but there was no interaction with the prompt conditions. Results from this study are compared with a previous study from our lab, and issues related to textbook familiarity, time limits, and plan generation skills are discussed.

T9. A Closer Examination of the Time-Dependent Effects of Benzodiazepines on Memory

Susan Jerrott, Sherry Stewart and Sandra Dunphy

Research has indicated that all benzodiazepines impair performance on tests of explicit memory while only lorazepam impairs performance on implicit memory tasks. However, Stewart, Rioux, Connolly, Dunphy & Teehan (1996) tested memory performance at two times after drug administration (2 hours and 2.8 hours) and found that at "time 2" both lorazepam and oxazepam impaired implicit memory performance. However, it remains unclear whether these novel findings represent true time-dependent effects of oxazepam or whether the implicit memory task was being contaminated at time 2 through subjects' use of explicit memory strategies. The present study was designed to clarify Stewart et al.'s findings. Thirty subjects were administered an acute dose of oxazepam, lorazepam or a placebo and were tested on a series of explicit and implicit memory tasks identical to those in the previous study. However, in this study subjects were only tested at Stewart et al.'s "time 2" in order to rule out the "implicit memory contamination" explanation of the previous findings. If oxazepam persists in producing impairments on the implicit memory test without any prior exposure to the explicit memory task, then the results will provide support for the "time-dependence" interpretation. The results of this study are currently being analyzed and will be presented and discussed.

T10. Management of Frequent Recurrent Headache: A Community Survey

P. McGrath, P. Forward, D. MacKinnon and J. Swann

This community telephone survey was designed to determine pain management patterns used by frequent headache sufferers. 1856 eligible households were contacted and 336 eligible contacts were identified. 274 frequent headache sufferers (81.6%) participated and describe their types of headaches, how they managed each and their attitudes to over the counter (OTC) medication. Headaches were classified using the International Headache Society's (IHS) system.

Females (n=33) reported an average of 1.9 types of headaches and males (n=141) reported 1.5 headache types. There were 465 headaches; 129 tension headaches, 158 migraine headaches, 8 chronic tension headaches and 148 headaches which were unclassifiable using IHS criteria. 56% of respondents used acetaminophen for tension type and 60% used acetaminophen for migraine. ASA was used by 15% for tension and 5% for migraine. Ibuprofen was used by 9% for tension and 8% for migraine. 9% used acetaminophen or ASA with 8mg of codeine for tension headache and 15% used these preparations for migraine. 1% used prescription medication for tension headache and 12% for migraine. The perceived effectiveness of OTC medication was approxiamtely 70% for tension headaches and 60% for migraine. Both tension headache and migraine headache sufferers waited about 1 hour before taking any medication. Tension headache sufferers waited until the headache was above 5 on a 0 to 10 scale (4.6 for migraine). Level of pain before taking medication was related to fear of tolerance and endorsement of stoicism. Fear of drug abuse predicted amount of medication taken.

Most community headache sufferers use OTC medication to manage headaches. They delay until pain is significant before taking any medication. Delay and amount of medication taken is related to attitudes to medication.

T11. Slow Down or You'll Miss It!

Sonya Symons, Heather MacLatchy-Gaudet, and Jacqueline Milner

Locating information in text is the most prevalent workplace literacy task, yet one that most people experience a lot of difficulty with. We know little about how to instruct students to locate information in text, primarily because we know little about the cognitive strategies that facilitate locating information. There are data suggesting that efficiency in locating information is associated with taking a planful approach to search, thereby allocating more time to locating the "best" places to find the information and less time to extracting information by reading text. Furthermore, reading rate data suggest that speed reading does not exist (i.e., that information, and even gist, is lost when readers increase their reading rate). We examined reading rate patterns of university students searching for information in a textbook chapter presented on a computer. We found that fast searchers differentiated between the pages of text that contained the targetted information (key pages) and those that did not (non-key pages) by reading the key pages at a slower rate than they read the non-key pages. Slower searchers skimmed both types of pages at similarly high rates. This increased the likelihood that they would miss the information and need to re-inpsect pages that they had already skimmed in order to locate the sought-after information. These data will be compared to normative reading rate data on skimming and scanning text in order to speculate on the interface of technolopgy and literacy.

T12. On Becoming a Person

John Barresi

What is a person? How does an animate or inanimate object become a person? In this talk I will consider the kinds of criteria we use to decide whether and when something becomes a person. I then consider and briefly discuss six ways for an object to become a person: 1) construction; 2) evolution; 3) development; 4) life-historical; 5) socio-cultural; and 6) cognitive and moral ideals. From a scientific point of view, each of these approaches can provide us with a deeper understanding of what it means to become a person.

T13. The Effect of Dopamine Antagonist Injected into the Ventral Tegmental Area on Self-Stimulation in Rats

Shi Ling and Shinshu Nakajima

It has been suggested that dopamine (DA) antagonists reduce the reinforcing effect of self-stimulation (SS) by acting on the terminal region of the mesolimbic DA pathway. There is a DA autoregulatory system in the ventral tegmental area(VTA) (A10). DA antagonists act on the autoreceptors on the soma and dendrites of DA neurons to block the inhibitory feedback to the cell body, causing a significant increase in cell firing. Further stimulation depolarizes A10 neurons to the point of acute depolarization block. Thus, there is a possibility that the depolarization block in the somatodendritic region of A10 neurons may also be responsible for the attenuation of SS after systemic injection of DA antagonist. Gamma-aminobutyric acid (GABA) is an inhibitory neurotransmitter which causes hyper-polarization. But under the situation of depolarization block, GABA should bring the depolarized neurons back to their resting state, restoring the ability to produce action potentials. To test this hypothesis, we injected DA antagonist raclopride alone or raclopride mixed with GABA into VTA. The results of preliminary tests show that injection of raclopride alone greatly attenuated SS, whereas injection of Raclopride-GABA mixture did not result in any suppression of SS. The results strongly support our hypothesis that the suppression of SS after systemic injection of DA antagonist may be caused by depolarization block in VTA.

T14. Stress and Ethanol Consumption in Rats

Tara L. Fidler and Vincent M. LoLordo

Two ideas have motivated research on the effects of stress on ethanol consumption in animals. The tension reduction hypothesis asserts that 1) ethanol reduces tension or anxiety and 2) animals will learn to consume ethanol for its tension reducing properties. The relief hypothesis asserts that the reinforcing effect of ethanol is enhanced in the presence of safety signals. There is some support in the literature for the relief hypothesis, but a series of experiments from our laboratory which included additional controls failed to support it. Experiments bearing on the tension reduction hypothesis are currently underway in our laboratory.

T15. Various Types of Linguistic Processings Studied with Event Related Brain Potential Measures: A Research Proposal

Myong Yoon and John Connolly

We plan to study charateristic changes in event-related potentials (ERP), recorded from normal subjects in response to the following types of linguistic processing tasks:
1) Semantic anomaly (between the main verb and its objective noun phrase, and/or between the subject and its main verb).
2) Various violations of local phrase structure rules (inversions between the main verb and its object NP, and/or between a preposition and its objective noun, faulty negative questions, etc.).
3) Faulty modifications of the matrix clause with subjacent clausal (infinitive or participle) phrases.
4) Negation and questions of semantically anomalous declaratives.
5) Bridging assumptions of antecedents.
6) Verification of an assertion about a state of affairs in the world.
7) Mental transformations for equivalent propositional functions.
8) Violation of quantity

T16. Error Patterns on the Continuous Performance Test in Unmedicated and Medicated Samples of Children With and Without ADHD: A Meta-Analytic Review

Bruno Losier, Pat McGrath and Ray Klein

Losier et al provide the first quantitative analysis of the performance deficits of children with ADHD on the Continuous Performance Test and the effects of methylphenidate on these deficits. Unmedicated samples of children with ADHD produce significantly more errors of omission and commission than their non-ADHD cohort. Moreover, methylphenidate reduce both types of errors in children with ADHD and, in the single study using them, in normal children as well. Additionally, submitting the data to the framework of the Signal Detection Theory suggest that children with ADHD are less sensitive to the difference between the targets and nontargets than non-ADHD children, while the two groups do not differ in response bias. Correspondingly, methylphenidate improves sensitivity while leaving bias unaffected.

T17. "I'd Rather be Wasted than Worried": Coping-Related Drinking in High AnxietySensitive Individuals

Alan MacDonald, Jan Baker and Sherry Stewart

Stewart and Pihl (1994) have shown that moderately intoxicating doses of alcohol reduce reactivity of highly anxiety sensitive subjects to signalled stressors compared to low anxiety sensitive controls. Based on these results, Stewart has suggested a modification of Sher's (1987) stress-response dampening theory to explain a heightened risk of alcohol abuse in high AS individuals (McNally, in press). The present study evaluated this theory by extending previous work on the sober reactivity of high AS vs. low AS subjects to a theoretically relevant stressor (i.e. hyperventilation, known to induce symptoms feared by high AS subjects) (Donnell & McNally, 1989), and the effects of alcohol administration on high AS vs. low AS subjects' responses to stress (Stewart & Pihl, 1994). Preliminary results suggest that high AS subjects administered a placebo showed greater reactivity to a hyperventilation challenge on affective and cognitive measures than low AS subjects. Furthermore, both high and low AS subjects who received a moderately intoxicating dose of alcohol showed less reactivity on somatic measures compared to placebo, but on affective and cognitive measures, the strongest reductions in reactivity were seen in high AS subjects. Thus, alcohol may be reinforcing for high AS subjects by eliminating their tendency to become fearful and catastrophize about aversive experiences.

T18. TBA

K. MacKinnon

T19. PTSD Following Terrorist Attacks: Situational Analysis, Prevalence, Risk Factors and Treatment

Yori (Yoram) Gidron, Ph.D.

Terrorist attacks (TA) consitute a growing source of trauma world-wide and particularly in Israel, and may result with post-traumatic stress-disorder (PTDS) in certain individuals. Several situational parameters are unique to TA (e.g., they are ideologically driven, affect a random sample of the population, are acute and "chronic" stressors). Approximately 26% of survivors (and rescuers) develop PTSD following TA. Certain situational parameters (degree of injury, degree of exposure) and personal parameters (e.g., prior depression) are significant risk factors for developing PTSD after TA. Group-therapy and debriefing are among the common treatment approaches, and provide opportunities for screening and treating survivors. These issues together with methodological limitations of previous studies will be discussed in this review, that is dedicated to the victims of terror in Israel and elsewhere.

T20. TBA

William C. Schmidt

T21. Age, Birth Order, and Emotional Coherence as Predictors of Performance on Theory of Mind Tasks

Doug Symons, Elizabeth McLaughlin and L. Noiles

birth order

Object identity and object location are two tasks designed to assess a child's theory of mind, or knowledge of the mental states of others. These tasks were administered to 70 children, ranging in age from 32 to 76 months (2 to 6 years). Consistent with previous research, performance on each task improved with age. The relative effectiveness of using age as a continuous, versus a categorical, variable will be discussed. In contrast to recently reported findings, birth order and number of siblings could only predict performance on one component of the object identity task (representational change). Children were also given an opportunity to identify emotional experience throughout each task by pointing to one of six photographs of faces exhibiting different emotional expressions. Preliminary analyses with 30 subjects revealed that children with more sophisticated emotional knowledge performed better on the theory of mind tasks, and that items answered appropriately had a more coherent emotional response than those addressed incorrectly. The data address the potential interdependence of the cognitive and emotional features of social understanding.

T22. What's the Point?

Chris Moore

orienting?

where do they look?

Infants start to point between 10 and 14 months of age. The pointing gesture, which seems to arise de novo in the child's behavioral repertoire at this time, has been described as an intentional communicative act which serves either to request an adult to get something (protoimperative) or to share with an adult attention to an interesting sight (proto-declarative). In both cases it is typically assumed that the point is used to redirect the attention of the adult towards something of interest. If so, then the infant must be aware of whether or not the adult has seen the target object or event. We have examined experimentally whether infants are sensitive to the adult's perspective by presenting the infant with a novel, interesting sight to the side when the mother is either looking at that sight, looking to the opposite side, or reading and not interacting. Spontaneous pointing was measured in all three conditions. In contrast to what common sense might suggest, infant of 12 and 18 months point more when mother has already seen the novel target than when her attention has been elsewhere. These results suggest that initially pointing does not serve to redirect another's attention but rather acts as a 'comment' on an already shared focus of attention.

T23. Self-awareness in Preschoolers: The Development of Temporal Self-continuity

Karen Skene

Self-continuity through time refers to a connection between the past, the present, and the future self. In this study, a group of 3-year-olds completed two tasks: a video self-recognition task to measure the past self and a delay of gratification task to measure the future self. Previous findings have indicated that older children are able to integrate their past and present selves as well as being able to choose greater rewards for their future self. Therefore, children who demonstrated self-recognition in the video task were expected to reliably choose the delayed reward in the future task. Results will be presented and issues for a second experiment will be discussed.

T24. The Ultrasonic Vocalization Paradigm as an Animal Model of Anxiety

Paul Mendella

Ultrasonic vocalizations (UVs) are emitted by rat pups and serve as a form of communication. For example, infant rats emit UVs in response to many situations including, separation from the litter, exposure to a cool environment, or tactile stimulation, and such calls elicit retrieval responses from the dam. Based on these findings, it is evident that UVs accompany a diverse range of ethologically relevant behaviours. Research has demonstrated the utility of the UV paradigm as a novel animal model of anxiety. In this respect, the UV paradigm has been revealed to be an effective method of screening novel non-benzodiazepine based anxiolytics. This finding is critically important as established animal models of anxiety have failed to externally validate the clinical effectiveness of non-benzodiazepine based anxiolytics. The vast majority of studies which have used the UV paradigm as an animal model of anxiety have focused on pharmacologically-induced alterations of UV rate only, and have not evaluated changes across other UV parameters (e.g., call frequency, call duration, time between calls). As a result, it is likely that many drug-induced changes in UV call patterns go unnoticed. In our lab, we now have the equipment necessary to concurrently measure different UV call parameters. Research attempting to determine the effects of various anxiolytic and anxiogenic agents across a wide range of UV call parameters will be discussed. The findings from these studies will help determine the utility of the UV paradigm as an animal model of anxiety.

T25. Teaching Via the WWW at Dalhousie: Are We Ready for Prime Time?

Ron Hoffman

A rapidly developing trend involves making teaching material available to students via the World Wide Web. This has a particular attraction for those interested in distance education, but is also useful to supply materials for regular university classes. Access to such material can be open and available to anyone in the world, or restricted to certain users, such as students registered in a particular class. This talk will demonstrate one such WWW site set up for Psychology 2500 this year, and point out some of the hazards encountered in trying to use this new technology.

HTML - Hypertext Mark Up Language.
- Tags inserted - used by Browser program
- add Links.

"World Lecture Hall"

T26. Webicising Your Survey, or How To Adminster Surveys over the World Wide Web

Hypertext Markup Language
Common Gateway Interface

Billy Schmidt

Survey Assistant

World Wide Web technology can act to decrease the amount of time, effort and costs incurred through survey research. Additionally, this medium opens one's subject pool to a mich wider, or specialized, audience than can be accessed through conventional channels. This talk will introduce the audience to the requirements, methods, and potential problems that are encountered when carrying out survey research on the World Wide Web.

T27. Auditory Gap Detection: On the Trail of the Processes Involved

T.L. Taylor, D.P. Phillips, S.E. Hall, M.M. Carr and J.E. Mossop

same as our other decision task. (Heather).

Speech sounds occur as a nearly continuous stream, broken only by short periods of silence (e.g., voice onset times). The durations of these "gaps" in the stream of sound are critical to the identification of some consonants. The perceptual processes involved in the discrimination of these silent intervals have typically been studied using an analytic "gap detection" paradigm, in which the task of the subject is to discriminate between two bursts of spectrally identical sound that differ only in that one of them contains a very brief silent interval ("gap"). We argue that the traditional gap detection paradigm is inappropriate to study the processes involved in speech recognition, because the gaps that occur in speech are bounded by spectrally dissimilar noises. We have redesigned the gap-detection paradigm to make it more speech-relevant, and have revealed two separable processes involved in the detection of silent intervals in ongoing streams of sound.

T28. More About Auditory Gap Detection: On the Trail of Voice Onset Times

Dennis Phillips, Tracey Taylor and Susan Hall

evolution of speech?

Voice onset times (VOTs) are the durations between the onset of the (broadband) consonantal burst and the onset of (low-frequency) voicing, and are what distinguish the acoustic signals for stop consonants of the same place of articulation (e.g., da and ta). VOT recognition is known to be categorical; studied with synthetic consonant-vowel syllables, subjects label all consonants as the short-VOT element unless the VOT exceeds a critical value (30-40ms), after which they label the consonant as the long-VOT element. The reasons for the categorical nature of the percept (and for the 30-40ms VOT cutoff) are not known, but its existence has been used as evidence that "speech is special". In the present experiment, we modified the traditional gap-detection paradigm so that we could study the processes involved in detecting VOT-like gaps in non-speech signals. Our data provide a new insight into why VOT discrimination is categorical, and into why the boundary between short and long perceived VOTs has the value that it does.

T29. Mental Rotation: It's All in the Name

Jeff Hamm and Paddy McMullen

The time to name objects rotated in the picture plane increases linearly as a function of the angular deviation from the upright for orientations between 0 and 120 degrees. Effects of orientation on naming latencies have been interpreted as support for orientation correction of misoriented images prior to object identification. After repeated presentations of stimuli, the magnitude of this effect of orientation tends to diminish. This diminishment is specific to a stimulus set. Alternative explanations suggest that orientation correction occurs after identification. Previous work has suggested that objects are initially identified at a basic categorical level (i.e. dog), and further visual processing is required for identification at a specific subordinate level (i.e. Collie). Orientation correction prior to identification predicts large effects of orientation for naming at both the basic and subordinate levels. Orientation correction after identification predicts minimal effects of orientation for naming at the basic level. Forty-eight subjects named two blocks of 72 line drawings depicting items from 6 basic categories. On the first presentation, subjects were instructed to give the basic level name while on the second block subjects gave the subordinate level name. Half the subjects were presented objects for a short exposure followed by a visual mask. The other subjects were presented objects until named. Results showed minimal effects of orientation for basic level naming with large effects with subordinate level naming. No effect of exposure duration was found with naming latencies. However, naming accuracy showed a significant linear decrease in the short exposure condition which did not interact with the level of identification. Results support the view that the majority of the effects of orientation are post-basic level identification.

T30. Magnocellular and Parvocellular Feature Manipulations in Visual Search

Sol Simpson and Paddy McMullen

According to Treisman's feature integration theory, searching for a conjunction of simple perceptual object features such as colour and form (e.g. a target red X among distractor blue Xs and red Os), requires attention and so results in a serial as opposed to a parallel search. Contrary to this prediction, McLeod, Driver and Crisp (1988) demonstrated that search for a moving X among moving Os and stationary Xs, pops out in a parallel manner. We speculated that this result may depend on a conjunction of features across separate processing channels such as the magnocellular (responsible for movement detection) and parvocellular (responsible for form detection) systems. To test this notion, a visual search task in which targets conjoined two magnocellular features, depth and direction of movement was designed. Stereo displays of dots moved in near and far depth planes. One plane contained a target that moved in one direction. In this plane, the target was embedded among distractors that moved in a different direction. Distractors in the other plane moved in both the same and different directions to the target. If parallel search results from a conjunction of features across channels, then this search should be serial. Alternatively, if search is parallel, whenever one magnocellular feature such as movement, needs to be conjoined with any other feature, this search should be parallel.

T31. Interocular Transfer of Motion Adaptation Effects: Higher-Order Processing of Motion Stimuli in Normal and Stereoanomalous Observers

Shelley McColl

Interocular transfer (IOT) of conventional motion aftereffects (MAE's) has been employed as a psychophysical technique to assess the binocularity of the human visual cortex. Like IOT of other aftereffects, conventional MAE's show partial IOT (approximately 60%). Recently, Raymond (1993) observed that with a higher-level motion coherence aftereffect, humans with normal stereopsis show complete IOT. Her argument is that conventional MAE's probe lower-level visual areas, where both monocular and binocular cells exist, whereas higher-order MAE's tap into higher-level visual areas, where cells are almost exclusively binocular. The present study revisited the issue of how stereoabilites are related to IOT using a higher-order motion task. The present results both support and extend Raymond's (1993) findings. Humans with normal stereoabilities showed partial transfer of a conventional MAE, whereas IOT was complete for the higher-order MAE. In contrast, observers with reduced stereopsis (including observers who were completely stereoblind) showed little, if any IOT of the conventional MAE, but showed high levels of IOT of the higher-order MAE. Implications of these findings for the neural mechanisms underlying the development of binocularity will be discussed.

T32. It's a Small World After All: Invertebrates as Model Systems for Studying the Physiology of Behaviour

Shelley Adamo

My laboratory is currently involved in three different research projects. 1) Parasitic effects on host behaviour. Parasites that alter host behaviour can be used as probes to help us understand how behaviour is regulated in normal animals. The tobacco hornworm, *Manduca sexta*, is parasitized by the wasp *Cotesia congregata*. During parasitism host feeding is suppressed, apparently by manipulating the host's octopaminergic system. 2) Psychoneuroimmunology of insects. Activating an insect's immune system can induce a change in its behaviour. In the cricket, bacteria induce behavioural fever, but gregarines and tachinids do not. Although raised temperatures increase survivial in crickets infected with bacteria, it has no effect on hosts infected with gregarines and tachinids. The latter 2 pathogens invoke a different type of immune response from that induced by bacteria. These different immunological mechanisms appear to have different effects on the central nervous system. 3) Communicating intent. Cephalopods such as the cuttlefish *Sepia officinalis*, visually signal their tendency to attack. a temperatures The lack of sexual dimorphism in this species may help to explain this unusual behaviour.

T33. Modulation by a Metabotrophic Glutamate Receptor Agonist of Phase Shifts Induced by VIP Injection Into the SCN

Paul Campagnaro, George Scott, and Benjamin Rusak

The suprachiasmatic nuclei (SCN) receive direct photic input via the retinohypothalamic tract (RHT), and glutamate is strongly implicated as the primary neurotransmitter of this pathway. Scott and Rusak (1996) found that 1S,3R-aminocyclypentane-1,3-dicarboxylic acid (ACPD), an agonist at type 1/5 metabotropic glutamate receptors (mGluR1/5), activates a large proportion of SCN neurons in vitro. These receptors are unusual among mGluR's in that they are positively coupled to phosphatidylinositol (PI) hydrolysis. Winder et al. (1993) reported that application of ACPD to rat hippocampal slices, potentiated cAMP formation stimulated by activating receptors coupled to adenylate cyclase, including vasoactive intestinal polypeptide (VIP) receptors. Further, Piggins et al. (1995) found that injection of VIP into the SCN region can phase shift hamster locomotor activity rhythms. We therefore investigated whether PI hydrolysis stimulated by mGluR1/5 activation can potentiate cAMP formation coupled to VIP activation in the SCN, and thereby modulate the phase shifting effects of VIP. Male Syrian hamsters were implanted with cannulas aimed at the SCN region and housed in constant dim red light, in cages equipped with running wheels monitored by a microcomputer. Animals were injected with VIP (150 pmol), ACPD (1.5 mM), VIP + ACPD (150 pmol & 1.5 mM), or saline vehicle (0.9%) at one of three circadian times (CT), CT6, CT13, or CT20 (where CT12 = activity onset). Injection of ACPD alone resulted in phase shifts which did not differ from vehicle injections. VIP injections produced small phase delays at CT13, advances at CT20 and no shifts at CT6. Coinjection of ACPD potentiated VIP-induced phase shifts selectively at CT13. These data support the hypothesis that IP hydrolysis coupled to mGluR1/5 activation can modulate the phase shifting effects of VIP receptor activation, early in the subjective night, possibly via potentiation of cAMP formation. (Supported by grants from the US AFOSR and NSERC of Canada)

T34. Gene Transfer Into The Rodent Nervous System

Savraj Grewal

The study of nerve cell death has become a major concern in neuroscience research over recent years. In particular, there has been a concerted effort to elucidate the cellular mechanisms of apoptosis, a distinct form of neuronal death. Extensive studies using isolated cells in vitro have identified 'families' of genes that appear to be important in defining the pathways leading to apoptotic cell death. Up- or down-regulation of these genes has been shown to both increase and decrease cell survival. However, a fundamental question is how relevent are these genes in modulating cell death in vivo. In this context, I will describe an approach we are taking in our laboratory to directly transfer genes to rat motor neurons in order to examine their effects on subsequent experimentally-induced neuronal damage.

T35. When the Lights are Turned Off, Something Stays Turned On

Mario Guido, Donna Goguen, Benjamin Rusak and Harold Robertson

The suprachiasmatic nucleus (SCN) of mammals is the site for the endogenous pacemaker that drives many biological rhythms. Light plays a key role in the mechanism of entrainment. Cells in the retinorecipient area of the SCN respond to retinal illumination with changes in both firing rates and immediate-early gene (IEG) expression. Fos and JunB are among the most studied IEGs; they exhibit a pronounced circadian variation in sensitivity to the inductive effects of light in the SCN, and they could be involved in mediating the response of the SCN to photic cues. We found that junB mRNA is expressed in the SCN of hamster in response to light throughout the subjective night and during the early subjective day (after 48 h in constant dark). Light-evoked expression was found throughout the SCN but was usually stronger in the ventral half. We were surprised to find, however, that junB mRNA was also expressed spontaneously in darkness during the transition from the projected night to the projected day. Its spontaneous expression began abruptly and persisted for at least 4 h into the projected day, and was always more robust in the dorsal SCN. Both the light-induced and spontaneous expressions of junB mRNA are translated into protein in SCN cells, with similar anatomical distributions. The expression of junB mRNA may play a role both in mediating circadian responses to photic stimuli and in spontaneous oscillation of elements of the SCN pacemaker.
(Supported by MRC of Canada and CONICET of Argentina).

T36. Is There an Effective Melatonin Receptor Antagonist?

Shui-Wang Ying, Benjamin Rusak, Philippe Delagrange, Elisabeth Mocaër, Pierre Renard and Béatrice Guardiola-Lemaître

We studied the effects of drugs related to the pineal hormone melatonin on neuronal firing activity in the suprachiasmatic nucleus (SCN), intergeniculate leaflet (IGL) and other brain areas in urethane-anesthetized Syrian hamsters. We tested melatonin and two naphthalenic derivatives of melatonin, a putative agonist (S20098) and a putative antagonist (S20928). Both melatonin and S20098, when given by systemically, were able to suppress firing rates of cells in a similar dose-dependent manner, but the effects of S20098 were longer lasting. Iontophoresis of melatonin dose-dependently depressed spontaneous and light-evoked activity of cells in the SCN and IGL, while iontophoresis of S20098 was relatively ineffective, probably because it is a poorly charged compound.

S20928 (2.0-10 mg/kg) alone decreased firing rates of light-sensitive cells the SCN and IGL; however, low doses (<2.0 mg/kg) of S20928 partially blocked the effects of melatonin agonists on most cells tested. The nonselective serotonin antagonist metergoline did not block the effects of either melatonin agonist, indicating that melatonin agonists did not act on serotonin receptors. Both melatonin agonists and antagonists were less effective when applied to cells in the hippocampus and dorsal lateral geniculate nucleus.

These results indicate that S20098 is an agonist acting probably on melatonin receptors in the Syrian hamster brain. S20928 may have mixed agonist/antagonist properties, but at low doses appears to function as an antagonist at melatonin receptors in the SCN and IGL, which are critical to the regulation of circadian rhythms.

P1. Illusory Line Motion, What Turns You On?

Ray Klein and John Christie

When a line is presented all at once with one end near the location of a peripheral stimulus or cue, the line appears to be drawn away from cue. We sought to determine if illusory line motion (ILM) is turned on by visual attention. Lines were presented as probes in the context of a detection task in which attention was endogenously or exogenously attracted to one of 2 circles by a central (arrow) or peripheral (brightening of one of the circles) cue. On most trials the cue was followed by a target (dot inside one circle) requiring a simple detection response. On the remaining trials, a line was presented, all at once or drawn over time, between the two circles and the subject indicated the direction and strength of perceived motion. Performance on the detection task revealed that attention was attracted to the cued location by both types of cues. ILM was strongly aroused by exogenous attention but was left unmoved by endogenous attention.

T20 Talk

P2. The Gradient Model of Illusory Line-Motion: Converging Support from Multiple Effects

William C. Schmidt

If a line appears all at once terminating nearby a previously cued spatial location, then motion within the line is experienced away from the cued location. This effect is known as illusory line motion. The explanation put forward by its discoverers is that the cue attracts attention, resulting in speeding signal transmission in the visual system along a spatial gradient extending from the cued location. This talk will present phenomenological problems for the gradient model and overcome them by extending the gradient model explanation to include the notion that the cue not only accelerates signal transmission, but extends the duration of such transmission along the gradient.

P3. What I Learned the Day-After the Infancy Conference

Dr. Doug Symons, Acadia University

In order to investigate interrelations of emotional adjustment, attachment security, and behavior problems, data from 57 mothers of 2-year-old infants was examined. Cluster analyses were conducted of maternal reports of emotional adjustment, parental stress, and coping. A three cluster solution resulted in three groups: 22 mothers with positive emotional adjustment, low parental stress, and high social support, 14 mothers with poor emotional adjustment, high parental stress, and low social support, and 21 mothers with high parental stress but good emotional adjustment. Attachment security was higher in group 1 than 2, and externalizing behavior problems were higher in groups 2 and 3 than 1. These data add to the literature suggesting maternal depression is related to infant attachment security in a normal sample, and further suggest behavior problems are related to attachment security. The data are the focus of a current paper being written.

P4. Non-invasive Predictors of Laterality in Speech Production and Comprehension

by Voyer, D.+, McGlone, J.*, and Savoie, J*.

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The present study investigated the value of non-invasive measures of laterality in predicting speech lateralization as determined with the intracarotid amobarbital procedure (IAP). Seventy right- and non-right-handed pre-operative persons with epilepsy completed four baseline laterality tasks (i.e., dichotic listening with words, tachistoscopic letter naming, object naming and dot localization). Verbal expression and comprehension were tested after the left and right injection of sodium amobarbitol. A multiple regression analysis indicated that handedness and dichotic listening laterality were significant predictors of both expression and comprehension, whereas object naming laterality was an additional predictor of expression. Differences between handedness groups were also obtained.

P5. More Somatic Than Cognitive-Affective Symptoms of Depression Among Individuals with Multiple Chemical Sensitivity

Anne-Marie Brown-DeGagne, Jeannette McGlone, and Darcy Santor

The Beck Depression Inventory (BDI) was used to document severity of depressive symptomatology in 42 individuals with Multiple Chemical Sensitivity (MCS). The purpose was to determine the extent to which somatic complaints contributed to the total BDI score. Analysis of cognitive-affective and somatic-performance complaints subscales indicated a significantly higher mean item score on somatic items relative to cognitive-affective items ($t = 6.43$, $p < .05$). The total BDI score classified a greater percentage (43%) of the sample as moderately depressed than did the cognitive-affective subscale score (29%). An item analysis of the BDI revealed that, relative to a sample of depressed outpatients, individuals with MCS tended to endorse more somatic items. Findings suggested that: a. the BDI total score overestimated severity of depressive symptomatology in this sample, and/or b. individuals with MCS tended to express depressive symptomatology in terms of somatic complaints. Conclusions recommended using caution to assess psychiatric disturbance in individuals with MCS with measures that include items of somatic complaints.